Bagadi Sarana Sai

Software Engineer | AI/ML | APIs | Data Pipelines
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PROFESSIONAL SUMMARY

Python Developer and recent Engineering graduate with hands-on experience building production-ready applications, REST APIs, and data-driven backends. Proficient in Python (OOP, DSA), FastAPI, Streamlit, and deployment with Docker and cloud platforms. Delivered four end-to-end Python projects spanning deep learning, NLP, and time-series forecasting. Passionate about clean, maintainable code and solving real-world problems with Python. Available immediately, open to relocation.

TECHNICAL SKILLS

Core Python	Python 3.x, OOP, DSA, Algorithms, Data Structures, Unit Testing
Web & APIs	FastAPI, Streamlit, REST APIs, HTTP, JSON, WebSockets, API Integration
Databases & ORMs	SQLite, SQL, FAISS, ChromaDB (Vector DBs), data modelling
Data & ML Libs	Pandas, NumPy, scikit-learn, TensorFlow, PyTorch, Keras, LangChain, Hugging Face
DevOps & Tools	Git, GitHub (CI), Docker (Basic), Render, Oracle Cloud, Jupyter, VS Code, Google Colab
Scripting & Automation	Python scripting, data pipelines, ETL, EDA, Feature Engineering, automation workflows
Soft Skills	Problem Solving, Quick Learner, Adaptable, Communication, Collaborative

PROJECTS

RAG-Based Smart Assistant GitHub Repo

- Built a production-ready Python application using LangChain, FAISS, and LLM APIs — covering document ingestion, chunking, embedding, and semantic retrieval.
- Designed a modular pipeline in Python with clean separation of concerns: loader, splitter, embedder, retriever, and response generator modules.

Pneumonia Detection API GitHub Repo | Live App

- Engineered a FastAPI backend serving a CNN model trained on 5,216 chest X-ray images; achieved 90%+ accuracy, F1 of 0.92, and ROC-AUC of 0.96. Deployed on Render cloud.
- Wrote Python data pipeline handling class-imbalance via computed class weights and augmentation; fine-tuned 265K of 7.3M parameters using transfer learning.

AI Load Forecasting — BiLSTM Streamlit App GitHub Repo | Live App

- Built an end-to-end Python time-series forecasting system: data ingestion → feature engineering → BiLSTM model training → interactive Streamlit dashboard.
- Engineered cyclical time features (sin/cos encodings), 24-hour rolling means, and sequence windowing using NumPy and Pandas; achieved R^2 of 0.9905. Deployed the Streamlit app as a live web service.

Customer Delinquency Prediction GitHub Repo

- Built a Python ML pipeline (XGBoost + Random Forest) for binary classification with thorough EDA, class-imbalance handling, and feature selection.
- Structured the project as a reusable Python package with config-driven preprocessing, model training, evaluation, and serialisation modules.

EXPERIENCE

Python / AI Intern -- Edunet Foundation | May 2025 – Jun 2025 | GitHub Repo | Live App | Credential

- Built and deployed an E-Waste Image Classification system using EfficientNetV2B2, TensorFlow/Keras, and Gradio UI, enabling real-time identification of 10 electronic waste categories via an interactive web application hosted on Hugging Face Spaces.
- Improved classification performance through transfer learning and image preprocessing techniques, achieving 96.67% accuracy, 96.71% precision, and 96.64% F1-score on a multi-class test dataset.

EDUCATION

B.Tech, Electrical and Electronics Engineering *2022 – 2026*
Rajiv Gandhi University of Knowledge Technologies, Ongole, AP | CGPA: 7.4

CERTIFICATIONS

Deloitte - Data Analytics Job Simulation on Forage-- Certificate Link
The Joy of Computing using Python course offered by IIT Ropar through NPTEL Portal – Certificate Link